

CENTREVILLE, AL, USA

WMO: 722290

Lat: 32.900N

Lon: 87.250W

Elev: 453

StdP: 14.46

Time Zone: -6.00 (NAC)

Period: 86-94

WBAN: 03881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.6	27.0	8.3	8.6	30.8	12.8	10.7	35.3	17.2	49.2	15.8	47.5	6.9	0	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	93.9	75.7	91.9	75.5	90.1	75.0	78.5	89.1	77.6	87.9	76.8	86.6	6.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	136.7	83.6	74.6	131.4	82.2	73.8	127.9	81.2	42.2	88.7	41.2	88.1	40.3	86.9	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.3	13.3	12.2	DB	14.2	97.7	6.1	1.0	9.8	98.5	6.2	99.1	2.8	99.7	-1.7	100.4
			WB	12.5	81.0	5.6	1.2	8.5	81.8	5.2	82.5	2.0	83.2	-2.1	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	45.6	50.1	56.2	62.9	70.3	77.1	80.1	79.4	74.6	62.8	55.2	47.4	(6)
	DBStd	14.11	8.35	9.63	8.92	7.53	5.24	3.60	2.79	3.88	5.67	7.31	8.84	9.83	(7)
	HDD50	559	183	106	42	7	0	0	0	0	0	5	50	166	(8)
	HDD65	2445	602	422	289	123	16	0	0	1	7	128	307	550	(9)
	CDD50	5500	46	108	236	393	629	813	933	913	738	401	206	84	(10)
	CDD65	1910	0	4	17	59	180	363	468	448	295	60	13	3	(11)
	CDH74	16723	0	12	173	591	1363	3255	4334	4176	2320	444	51	4	(12)
	CDH80	6671	0	0	12	115	356	1362	1987	1887	857	94	1	0	(13)
(14) Wind	WSAvg	6.3	7.0	7.5	7.6	6.9	6.0	5.3	5.0	5.1	5.6	6.2	7.0	7.1	(14)
(15) Precipitation	PrecAvg	54.0	5.3	5.5	5.5	4.4	3.4	4.3	4.8	3.9	4.0	2.8	4.4	4.5	(15)
	PrecMax	63.2	10.0	10.7	9.8	8.5	8.6	9.0	10.7	6.2	8.3	8.7	8.6	7.6	(16)
	PrecMin	40.7	2.5	1.7	1.4	1.7	0.2	1.3	1.0	1.2	0.9	0.0	1.6	2.8	(17)
	PrecStd	5.7	2.2	2.2	2.2	1.9	2.2	2.2	2.5	1.4	2.0	1.9	1.9	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.4	76.0	81.9	86.9	88.1	94.3	96.9	96.6	93.0	87.3	78.5	73.5	(19)
		MCWB	59.5	65.6	63.3	66.5	70.7	73.7	77.2	75.9	74.7	71.5	65.0	67.5	(20)
	2%	DB	66.0	71.4	78.4	83.2	86.3	91.7	94.1	93.8	90.1	82.3	75.2	69.2	(21)
		MCWB	57.4	60.8	60.9	65.4	70.4	73.5	76.4	75.8	74.1	67.8	63.8	63.9	(22)
	5%	DB	62.5	68.1	75.6	80.4	84.1	89.6	91.9	91.6	87.7	79.0	72.3	65.9	(23)
		MCWB	55.6	59.3	60.6	64.6	69.8	73.5	76.4	75.5	73.6	65.6	62.7	59.7	(24)
	10%	DB	59.4	65.0	71.9	77.4	81.7	87.5	89.8	89.4	85.3	75.6	69.4	62.5	(25)
		MCWB	53.7	58.1	59.2	62.9	68.8	73.2	76.0	75.3	73.1	63.7	62.2	57.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	67.1	69.1	71.5	75.3	78.2	80.4	79.8	78.2	74.3	70.6	68.5	(27)
		MCDB	67.4	73.3	73.7	78.7	83.9	87.9	91.4	90.0	86.7	83.3	74.7	72.0	(28)
	2%	WB	60.6	64.8	65.7	69.6	73.8	76.8	78.9	78.3	76.8	71.6	68.2	65.3	(29)
		MCDB	64.3	68.9	72.6	77.1	81.6	86.6	90.2	89.0	86.1	77.2	71.4	68.4	(30)
	5%	WB	57.9	62.7	63.5	67.5	72.3	75.8	77.9	77.3	75.7	69.2	66.0	61.9	(31)
		MCDB	60.9	66.4	71.6	76.2	79.6	85.7	88.5	88.0	84.4	75.2	69.8	64.8	(32)
	10%	WB	54.1	59.3	61.1	65.6	70.9	74.9	77.0	76.3	74.6	67.0	63.7	58.6	(33)
		MCDB	57.6	63.4	68.4	75.1	78.0	84.5	86.9	86.2	82.5	72.9	67.6	61.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	19.7	21.7	22.9	19.5	19.6	18.9	19.1	18.7	21.5	20.4	18.0	(35)
		MCDBR	23.1	21.8	25.9	26.6	21.7	22.5	22.0	22.0	20.8	23.5	21.8	19.6	(36)
	5% WB	MCWBR	16.4	13.7	13.3	12.4	9.2	9.0	8.2	7.4	8.9	11.8	13.9	15.0	(37)
		MCWBR	17.7	17.3	19.9	21.0	17.5	19.8	20.0	19.5	18.2	18.1	16.3	16.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.319	0.329	0.358	0.399	0.451	0.482	0.540	0.536	0.443	0.368	0.342	0.321	(40)
		taud	2.531	2.501	2.435	2.302	2.204	2.154	2.007	2.014	2.261	2.461	2.477	2.540	(41)
	Ebn at Noon		286	294	292	283	267	258	242	241	261	276	274	279	(42)
		Edh at Noon	27	30	35	42	46	48	56	55	41	31	28	25	(43)
(44) All-Sky Solar Radiation	RadAvg		856	1030	1394	1761	1940	1985	1911	1764	1560	1304	970	732	(44)
	RadStd		65	135	129	132	140	154	125	104	132	160	99	75	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.94	N/A	N/A	N/A	N/A	N/A	N/A	+277	+193

Nomenclature: See separate page